DevOps Lab Assessment - Experiment 9

Master Lab Record Book: All Dockerfiles, Manifests, Commands & Observations

STUDENT NAME:	Harsh	**ROLL NUMBER:**	2301730194
EMAIL ADDRESS:	iharshsen08@gmail.com	**ASSESSMENT:**	Question 9 (5 Marks)
COURSE / LAB:	DevOps Lab 2 (Containerization & K8s)	**ENVIRONMENT:**	Docker 29 / Minikube v1.36 / kubectl

1. Compendium Overview & Candidate Verification

Candidate: Harsh | Roll No: 2301730194 | Email: iharshsen08@gmail.com. This compendium serves as the comprehensive master record of all laboratory artifacts, configurations, declarative manifests, automation workflows, commands, and empirical observations developed throughout Lab 2 (Containerization & Kubernetes Orchestration). It centralizes all code and operational playbooks for auditing and archival in the student repository.

Exp #	Laboratory Topic	Key Artifacts Produced	Status / Outcome
1	K8s Deployment & Service	deployment.yaml, service.yaml, configmap.yaml	Verified 3 Pods & NodePort
2	Rolling Updates & Scaling	ReplicaSet scaling (3 to 5 to 2), rollout undo	Zero downtime confirmed
3	Comprehensive Technical Report	10-mark formal technical report & architecture	Complete system evaluation
4	Docker Containerization	Dockerfile (Multi-Stage), app.py, requirements	142 MB image, UID 10001
5	Docker Management & Compose	docker-compose.yml (Web + Redis + Vol + Net)	Multi-container state synced
6	Docker Storage Persistence	run_storage_demo.sh, named volumes, bind mounts	Data loss & persistence verified
7	Docker Networking & DNS	run_network_demo.sh, custom bridge net	DNS (127.0.0.11) resolution
8	Continuous Deployment CI/CD	.github/workflows/deploy.yml, Kustomize	Automated GitOps CD

2. Declarative Code Repository Summary

```
// File: src/app/Dockerfile | Candidate: Harsh (2301730194)

# --- Multi-Stage Production Dockerfile (Candidate: Harsh) --- FROM python:3.11-slim AS builder WORKDIR
/app COPY
requirements.txt . RUN pip install --no-cache-dir --user -r requirements.txt FROM python:3.11-slim AS
runner WORKDIR /app
RUN groupadd -g 10001 appgroup && useradd -u 10001 -g appgroup -s /bin/false appuser COPY --from=builder
/root/.local
/home/appuser/.local COPY app.py . RUN chown -R appuser:appgroup /app /home/appuser USER appuser EXPOSE
5000 CMD
["gunicorn", "--bind", "0.0.0.0:5000", "--workers", "3", "app:app"]
```

3. Master DevOps CLI Command Reference

Domain	Command Syntax	Operational Purpose
Docker Build	docker build -t app:v1 -f Dockerfile .	Builds OCI image from Dockerfile context.
Docker Run	docker run -d -p 5000:5000 --name c1 app:v1	Runs background container with port forward.
Docker Volumes	docker run -v vol1:/data alpine	Mounts persistent named volume.
Docker Network	docker network create --driver bridge net1	Creates custom network with embedded DNS.
Docker Compose	docker compose up -d / down -v	Spins up or destroys entire multi-container stack.
K8s Apply	kubectl apply -f manifest.yaml	Declarative resource creation/update.
K8s Scaling	kubectl scale deployment/app --replicas=5	Horizontally scales pod instances.
K8s Rollout	kubectl rollout status / undo deployment/app	Tracks progress or reverts faulty revision.

4. Comprehensive Summary & Conclusion

Master Record Sign-Off

• Master Record Sign-Off: Completed by candidate Harsh (Roll No: 2301730194, Email: iharshsen08@gmail.com).

• This master record book encapsulates all requirements of DevOps Lab 2. All container configurations, Kubernetes manifests, and automated CI/CD pipelines adhere to modern cloud-native standards (immutability, least-privilege execution, zero-downtime rolling upgrades, persistent storage isolation, and automated self-healing).